

High-performance MCU with RISC-V single-core and dual-core microprocessors
Powerful Image and voice processing capability 10 MB or 32 MB PSRAM in the chip's package
55 GPIOs, rich set of peripherals QFN104 (10x10 mm) Package

USB 2.0 OTG high-speed interface (DW/DP)

ESP32-PinHeader (QFN104)

Table 2-4. Boot Mode Control

Boot Mode	GPIO35	GPIO36	GPIO37	GPIO38
SPI Boot Mode (default)	1	x ¹	x ¹	x
UART Download Boot mode ²	0	0	1	x
SPI Download Boot mode ³	0	0	1	x
Invalid Combination ⁴	0	0	0	0

[illegible]

The power consumption of microSD cards varies by manufacturer, but appears to be in the range of 40-100 mA (20-100 mA at a supply voltage of 3.3 V). Specifications from Toshiba microelectronics list a maximum of 140 mA (40 mA during transfer).

Left Diagram: Full Connection

Power: +3.3V to VDDIO (1.8V/3.3V) ESP_ID0_V04. VDDIO is connected to R15 (4.7k/R0402) and R16 (22k/R0402). GND is connected to R14 (4.7k/R0402) and R17 (22k/R0402).

MicroSD Card: R18 (4.7k/R0402) to SD1_D3, R19 (4.7k/R0402) to SD1_CMD, R20 (4.7k/R0402) to SD1_VDD, R21 (4.7k/R0402) to SD1_CLK, R22 (4.7k/R0402) to SD1_D1, R23 (4.7k/R0402) to SD1_D2, R24 (4.7k/R0402) to SD1_D0.

Control: R25 (4.7k/R0402) to GND, R26 (4.7k/R0402) to GND, R27 (4.7k/R0402) to GND, R28 (4.7k/R0402) to GND, R29 (4.7k/R0402) to GND, R30 (4.7k/R0402) to GND, R31 (4.7k/R0402) to GND, R32 (4.7k/R0402) to GND, R33 (4.7k/R0402) to GND, R34 (4.7k/R0402) to GND, R35 (4.7k/R0402) to GND, R36 (4.7k/R0402) to GND, R37 (4.7k/R0402) to GND, R38 (4.7k/R0402) to GND, R39 (4.7k/R0402) to GND, R40 (4.7k/R0402) to GND, R41 (4.7k/R0402) to GND, R42 (4.7k/R0402) to GND, R43 (4.7k/R0402) to GND, R44 (4.7k/R0402) to GND, R45 (4.7k/R0402) to GND, R46 (4.7k/R0402) to GND, R47 (4.7k/R0402) to GND, R48 (4.7k/R0402) to GND, R49 (4.7k/R0402) to GND, R50 (4.7k/R0402) to GND, R51 (4.7k/R0402) to GND, R52 (4.7k/R0402) to GND, R53 (4.7k/R0402) to GND, R54 (4.7k/R0402) to GND, R55 (4.7k/R0402) to GND, R56 (4.7k/R0402) to GND, R57 (4.7k/R0402) to GND, R58 (4.7k/R0402) to GND, R59 (4.7k/R0402) to GND, R60 (4.7k/R0402) to GND, R61 (4.7k/R0402) to GND, R62 (4.7k/R0402) to GND, R63 (4.7k/R0402) to GND, R64 (4.7k/R0402) to GND, R65 (4.7k/R0402) to GND, R66 (4.7k/R0402) to GND, R67 (4.7k/R0402) to GND, R68 (4.7k/R0402) to GND, R69 (4.7k/R0402) to GND, R70 (4.7k/R0402) to GND, R71 (4.7k/R0402) to GND, R72 (4.7k/R0402) to GND, R73 (4.7k/R0402) to GND, R74 (4.7k/R0402) to GND, R75 (4.7k/R0402) to GND, R76 (4.7k/R0402) to GND, R77 (4.7k/R0402) to GND, R78 (4.7k/R0402) to GND, R79 (4.7k/R0402) to GND, R80 (4.7k/R0402) to GND, R81 (4.7k/R0402) to GND, R82 (4.7k/R0402) to GND, R83 (4.7k/R0402) to GND, R84 (4.7k/R0402) to GND, R85 (4.7k/R0402) to GND, R86 (4.7k/R0402) to GND, R87 (4.7k/R0402) to GND, R88 (4.7k/R0402) to GND, R89 (4.7k/R0402) to GND, R90 (4.7k/R0402) to GND, R91 (4.7k/R0402) to GND, R92 (4.7k/R0402) to GND, R93 (4.7k/R0402) to GND, R94 (4.7k/R0402) to GND, R95 (4.7k/R0402) to GND, R96 (4.7k/R0402) to GND, R97 (4.7k/R0402) to GND, R98 (4.7k/R0402) to GND, R99 (4.7k/R0402) to GND, R100 (4.7k/R0402) to GND.

Middle Diagram: p1EXT1 Header

Power: +3.3V to VDDIO (1.8V/3.3V) ESP_ID0_V04. GND is connected to R14 (4.7k/R0402) and R17 (22k/R0402).

MicroSD Card: R18 (4.7k/R0402) to SD1_D3, R19 (4.7k/R0402) to SD1_CMD, R20 (4.7k/R0402) to SD1_VDD, R21 (4.7k/R0402) to SD1_CLK, R22 (4.7k/R0402) to SD1_D1, R23 (4.7k/R0402) to SD1_D2, R24 (4.7k/R0402) to SD1_D0.

Control: R25 (4.7k/R0402) to GND, R26 (4.7k/R0402) to GND, R27 (4.7k/R0402) to GND, R28 (4.7k/R0402) to GND, R29 (4.7k/R0402) to GND, R30 (4.7k/R0402) to GND, R31 (4.7k/R0402) to GND, R32 (4.7k/R0402) to GND, R33 (4.7k/R0402) to GND, R34 (4.7k/R0402) to GND, R35 (4.7k/R0402) to GND, R36 (4.7k/R0402) to GND, R37 (4.7k/R0402) to GND, R38 (4.7k/R0402) to GND, R39 (4.7k/R0402) to GND, R40 (4.7k/R0402) to GND, R41 (4.7k/R0402) to GND, R42 (4.7k/R0402) to GND, R43 (4.7k/R0402) to GND, R44 (4.7k/R0402) to GND, R45 (4.7k/R0402) to GND, R46 (4.7k/R0402) to GND, R47 (4.7k/R0402) to GND, R48 (4.7k/R0402) to GND, R49 (4.7k/R0402) to GND, R50 (4.7k/R0402) to GND, R51 (4.7k/R0402) to GND, R52 (4.7k/R0402) to GND, R53 (4.7k/R0402) to GND, R54 (4.7k/R0402) to GND, R55 (4.7k/R0402) to GND, R56 (4.7k/R0402) to GND, R57 (4.7k/R0402) to GND, R58 (4.7k/R0402) to GND, R59 (4.7k/R0402) to GND, R60 (4.7k/R0402) to GND, R61 (4.7k/R0402) to GND, R62 (4.7k/R0402) to GND, R63 (4.7k/R0402) to GND, R64 (4.7k/R0402) to GND, R65 (4.7k/R0402) to GND, R66 (4.7k/R0402) to GND, R67 (4.7k/R0402) to GND, R68 (4.7k/R0402) to GND, R69 (4.7k/R0402) to GND, R70 (4.7k/R0402) to GND, R71 (4.7k/R0402) to GND, R72 (4.7k/R0402) to GND, R73 (4.7k/R0402) to GND, R74 (4.7k/R0402) to GND, R75 (4.7k/R0402) to GND, R76 (4.7k/R0402) to GND, R77 (4.7k/R0402) to GND, R78 (4.7k/R0402) to GND, R79 (4.7k/R0402) to GND, R80 (4.7k/R0402) to GND, R81 (4.7k/R0402) to GND, R82 (4.7k/R0402) to GND, R83 (4.7k/R0402) to GND, R84 (4.7k/R0402) to GND, R85 (4.7k/R0402) to GND, R86 (4.7k/R0402) to GND, R87 (4.7k/R0402) to GND, R88 (4.7k/R0402) to GND, R89 (4.7k/R0402) to GND, R90 (4.7k/R0402) to GND, R91 (4.7k/R0402) to GND, R92 (4.7k/R0402) to GND, R93 (4.7k/R0402) to GND, R94 (4.7k/R0402) to GND, R95 (4.7k/R0402) to GND, R96 (4.7k/R0402) to GND, R97 (4.7k/R0402) to GND, R98 (4.7k/R0402) to GND, R99 (4.7k/R0402) to GND, R100 (4.7k/R0402) to GND.

Right Diagram: p1EXT1 Header

Power: +3.3V to VDDIO (1.8V/3.3V) ESP_ID0_V04. GND is connected to R14 (4.7k/R0402) and R17 (22k/R0402).

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